6-1.1, (a) (b), 6-1.2, 6-3.1, 6-3.2 (b) (c), 6-5.3 (a) (c)

1. A random process having sample function of the form $X(t) = A + B\sin(\omega t + \theta)$, where A is an exponential RV with parameter $\lambda = 3$, B is a Gaussian RV with mean 3 and variance 4, ω is uniformly distributed between 100 and 110, and θ is uniformly distributed between 0 and 2π. All the RVs are mutually independent. Is the random process mean ergodic?

2. In the above problem, if A is a constant instead of a RV, is the random process mean ergodic?